Arabic NLP Open Source: A-to-Z Documentation

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1. The Why: The Problem and Our Solution

The Challenge

The Arabic NLP ecosystem suffers from a severe scarcity of open-source, ready-to-use datasets compared to English. High-quality, evaluated models for local dialects are even rarer. This vacuum significantly slows down the development of AI applications tailored for Arabic speakers.

The Solution

Our mission is to build a foundational open-source Arabic contribution in the NLP field. This involves collecting and cleaning a massive dataset, developing text processing tools, and publishing the results on GitHub and HuggingFace Hub with professional open-source standards—creating a reliable reference for the global tech community.

2. The What: The Product & MVP Scope

This project is defined by three core deliverables aimed at researchers, NLP developers, and the broader open-source community:

- **Arabic NLP Dataset:** A high-quality, documented dataset of over 10,000 texts. It is meticulously annotated for Topic Classification, complete with a comprehensive Data Card, and published on HuggingFace Hub under GDSC KSU.
- **Arabic NLP Pipeline:** A dedicated preprocessing pipeline utilizing spaCy and NLTK (covering Tokenization, Normalization, and Stopword Removal), alongside a baseline Text Classification model trained directly on our dataset.
- **Robust CI/CD & Documentation:** Automated GitHub Actions pipelines for continuous testing and linting, professional dual-language documentation (Arabic/English), and a detailed CONTRIBUTING.md to foster external contributions.

3. The How: Annotation & Technical Architecture

Annotation Specifications (Topic Classification)

To ensure high-signal data, texts are exclusively classified into one of six core categories based on actual content:

Label (Code)	Description & Examples
politics	Government affairs, international relations, elections, political decisions.
sports	Football, Olympics, championships, athletic achievements.
technology	AI, software, inventions, scientific research.
culture	Arts, literature, heritage, cultural festivals, cinema.
economy	Financial markets, commerce, corporations, budgets, economic growth.
health	Diseases, medical research, healthcare, medicines, health awareness.

Strict Dataset Constraints

- **Accepted:** Modern Standard Arabic (Fusha) ONLY. Minimum length of 50 words per text.
- **Rejected:** Any local dialects (Saudi or otherwise), entirely religious content, or excessively short texts.

The Step-by-Step Labeling Pipeline

1. Contributors are assigned specific textual batches to prevent overlap.
2. Labels are assigned manually via structured Google Sheets (Text | Label | Annotator).
3. Upon reaching 2,000 manually annotated texts, an automated pre-classification model is integrated to assist and accelerate the process.
4. A mandatory **10% randomized quality assurance review** is conducted by the Project Lead to ensure strict adherence to the guidelines.

Technical Stack

- **Language:** Python 3.10+
- **NLP Framework:** spaCy + NLTK (Preprocessing Pipeline)
- **Language Models:** HuggingFace Transformers (Text Classification & Embeddings)
- **Infrastructure:** HuggingFace Hub (Dataset hosting), GitHub Actions (CI/CD)